ISCED: 0612
1. Course Code: CSE-3634
2. Course Title: Computer Networks Lab
3. Credit Hours: 1.5
4. Contact Hours: 3 lab hours per week
5. Type: Core, Engineering
6. Prerequisite: CSE-3525 (Data Communication)
7. Co-requisite: CSE-3633 (Computer Networks)
8. Instructor, Class Schedule and Locations:
Instructor: Abdullahil Kafi
Class Hours: As per Class Schedule

9. Course Rationale / Summary: The subject of computer networking is enormously complex, involving many concepts, protocols, and technologies. To cope with the scope and complexity these protocols and technologies are woven together in an intricate manner in what is called the layered protocol stack (or suite). The layered organization allows breaking down complex functions required for computer networking into manageable tasks. The course uses the Internet's architecture and protocols as the primary vehicle for studying fundamental computer networking concepts. More than often, the course will also include concepts and protocols from other network architectures. But the main focus is on the Internet, a fact reflected in organizing the course around the Internet's five-layer architecture. Computer networks and the Internet have grown from a research curiosity to something as essential as the ubiquitous telephone and utility networks. It has been able to withstand rapid growth fairly well and its core protocols have been robust enough to accommodate applications (e.g., the World Wide Web) that were unforeseen by the original Internet designers. Networking is becoming an essential component of many systems.

10. Course Objective:
a. The greater goal of this course in to introduce students the advance software tool for doing research as well as to enable them doing elementary works in the industry of computer networks
b. Analyze the performance of popular link layer protocols to match the results with that of the theory
c. Introducing simulation tools to show and implement the way existing communication protocols work
d. Demonstrate some of the communication protocols at work

11. Course Learning Outcomes (CLOs):
Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	Demonstrate a fair command over Discrete Event Network simulator for performance analysis.	30%
2.	Apply fair command over Network simulator for configuring different types of Networks.	40%

3	Implementing and Investigating the existing communication protocols to improve the efficiency of those protocols	30%

12. Course Learning Outcomes (CLOs) and Mapping:

CLO	CLO Statement	PLO	DL	KP	EP	EA	Delivery methods and activities	Assessment Tools
1	Demonstrate a fair command over Discrete Event Network simulator for configuring different types of Networks and	5	Cognitive/Apply	6			Lecture, Class discussion, Assignment, Lab work, Project work.	Performance Report writing Teamwork Viva Presentation
2	Implementing and Investigating the existing communication protocols to improve the efficiency of those protocols.	4	Cognitive/Create	3,4,5			Lecture, Class discussion, Assignment, Lab work, Project work.	Performance Report writing Teamwork Viva Presentation
3	Design and Developing elementary things that might need in future research in the telecommunication industry.	3	Cognitive /Create	8		1	Lecture, Class discussion, Assignment, Lab work, Project work.	Performance Report writing Teamwork Viva Presentation
Note: DL: Domain/level of learning taxonomy, **KP:** Knowledge Profile, **EP:** Attribute of Complex Engineering Problems, **EA:** Attribute of Complex Engineering Activities **Learning Domains** (C: Cognitive, A: Affective, P: Psychomotor)								

CO/PO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					1							
CLO2				1								
C?O3			1									

14. Resources:

1. Computers with minimum 8 GB RAM
2. OMNet++ Discrete Event Simulator
3. GNS3 or CISCO packet Tracer simulation software

15. Text Books:

#	Authors	Title of Book Edition	Publishing Year	ISBN
1.	András Varga	Omnet++ Simulation manual Version 5.5	2016	
2.	Team Contribution	INET Framework for OMNeT++ manual	2016	
3.	Todd Lammle	Cisco Certified Network Associate study guide 6th edition	Wiley Publishing Inc. 2007	ISBN-13: 978-0-470-11008-9

Online Resources:
I) https://omnetpp.org/
II) https://inet.omnetpp.org/
iii) https://docs.gns3.com/docs/

16. Weightage Distribution among Assessment Tools:

Assessment Tools	Weightage (%)
Punctuality	10
Continuous Assessment	60
Team Project	30

17. Grading Policy:
As per IIUC grading policy

18. Weekly Activity Plan and List of Experiments:

Week	Activities	Experiments
1	Lab work	Install and introduce different parts of a network simulator. Implementing a simple network of two nodes using the simulator.

2	Lab work	Enhancing the two-node network with additional features. Develop the simple network into a simple real-world network.
3	Lab work	Analyze results (throughput, delay etc.) of the simple network.
4	Lab work	Develop the simple network into a simple real-world network.
5	Lab work	Implement pure ALOHA using the simulator and measure its performance
6	Lab work	Implement slotted ALOHA using the simulator and measure its performance
7	Lab work	Implement IEEE 802.11 using the simulator and measure its performance
8	Lab work	Study of different types of Network cables and practically implement the cross-wired cable and straight through cable using a crimping tool.
9	Lab work	Introducing GNS3 software and study of basic commands of networked devices and network configuration commands.
10	Lab work	Construct a simple LAN using packet tracer; apply VLSM in IP planning.
11	Lab work	Configure and analysis a network using Distance Vector Routing Protocol
12	Lab work	Configure and analyze a network using Link State Routing Protocol.
13	Lab work	Configure and analyze a network using Adhoc Routing Protocol Team Project works.
14	Project Work	Open ended capstone project by the team.

19. Rubrics

Rubrics of Labworks and Projects						
Course: Computer Networks Lab			Course Code: CSE3634		Instructor: Abdullahil Kafi	
Lab Activity	**Marks**	**Poor**	**Unsatistory**	**Satisfactory**	**Good**	**Excellent**
1. Mastering over the tools, commands and simulation software.	20	Unable to use necessary tools and commands.	Able to use some but fail to modify.	Able to use tools and commands but fail to explain properly.	Able to use tools and commands with clear understanding.	Able to use the tools and commands with efficiency.
2.Mastering	20	Unable to	Able presents	Able to	Able to	Able to

on Network analysis		present and analyze the networks' data.	data but fails to interpret.	present and analyze some but not all.	present and analyze network data.	present and analyze network data with clear understanding.
3. Mastering on Network protocols	20	Unable to explain and implement network protocols.	Able to explain but unable to implement network protocols.	Able to explain and implement network protocols partially.	Able to explain and implement network protocols correctly.	Able to explain and implement network protocols efficiently..
4. Uniqueness and genuinity on ideas and implementation techniques.	20	No self contribution	No self contribution	Have some self contribution	Have good contribution	Have excellent contribution.
5. Report and Presentation	20	Required features are missing in report and presentation.	Required features are missing either in report or in presentation.	Required features are included in both report and presentation.	Required features are included in both report and presentation accurately	Report and presentation contains both required features as well as extra features.
Total	100					

20. List of Project(Open Ended):

Project Title
1 Improving/adding routing, DNS and NAT on VLSM Campus Network 0to255 2 Implementation of Static NAT on Campus Area Network of VLSM Barbarians 3 Enhancing the Campus Network Security by implementing OSPF Authentication Error404 4 ACL based Campus Network Anonymous 2.0 5 Extend Campus Network by adding Security features in GNS3 Version 4.0 6 Static Routing with VLSM Campus Network 7 OSPF Routing with VLSM Campus Network 8 BGP Routing with VLSM Campus Network

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